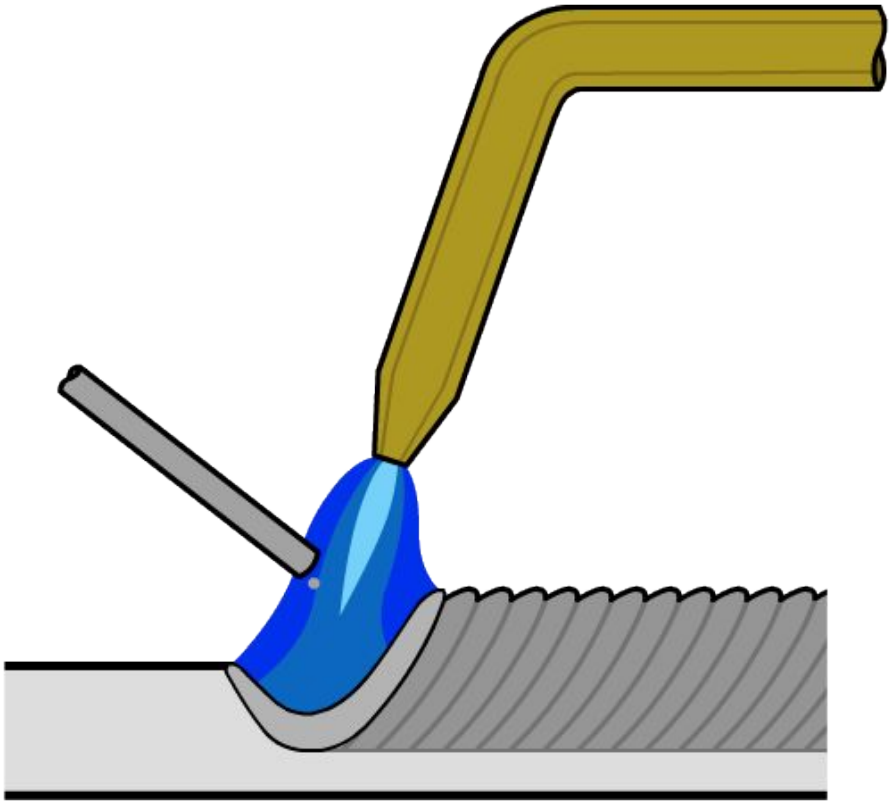
Mr Dimelow

3 Processes, techniques and specialist tools

3.4 Permanent and Semi-Permanent Joining Techniques

c) heat – oxy-acetylene welding, MIG welding, brazing, hard soldering, soft soldering (including use of specialist tools)

Gas Welding



In this presentation you will:

- explore the gas welding process, including safety concerns

Introduction

In gas welding the heat is produced by burning a fuel gas with oxygen.

It was once the main welding process for fabrication and construction.

Today, its use is limited to specialist purposes.



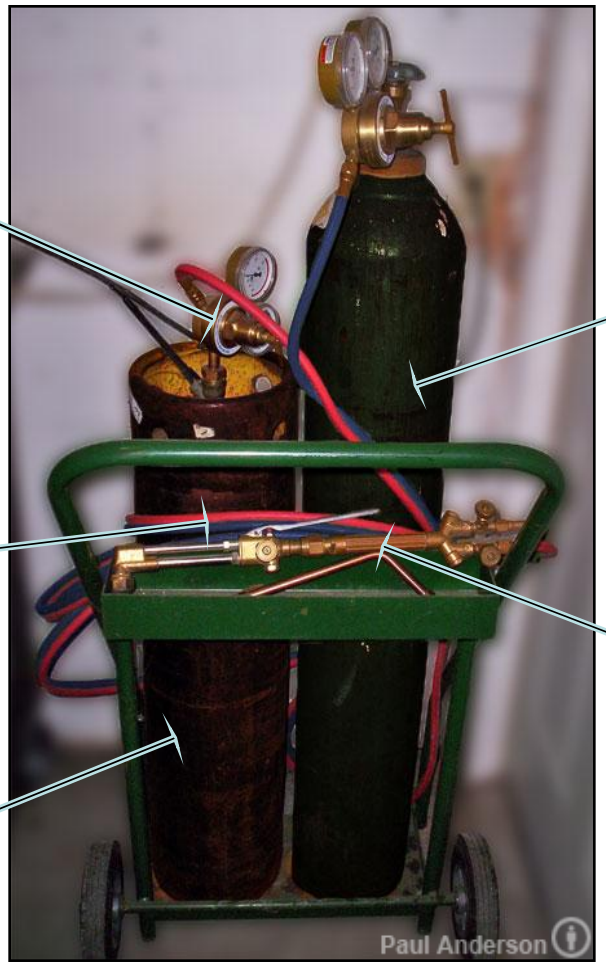


Next >

Gas Welding Equipment

These are the components of a gas welding rig:

- *Fuel and oxygen gas cylinders*
- *Valves, pressure regulators and gauges*
- *Colour coded gas hoses*
- *Welding torch*



Next >

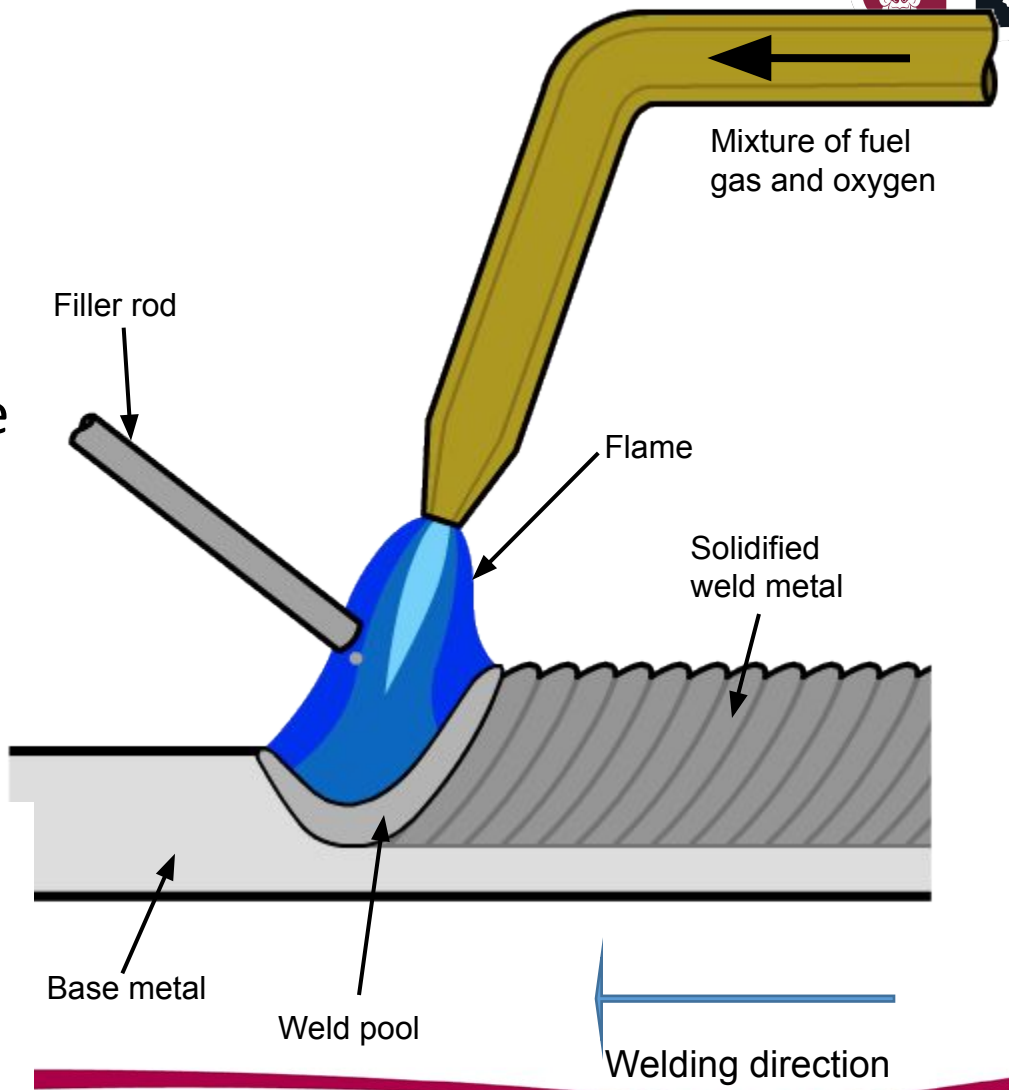
The Welding Process

The welding torch burns a fuel gas with oxygen to produce a high-temperature flame.

The heat melts both the base metal and a metal filler rod.

The molten metals flow together, then solidify.

The filler rod contains flux that vaporises to displace air and stop oxide contamination.



Next >

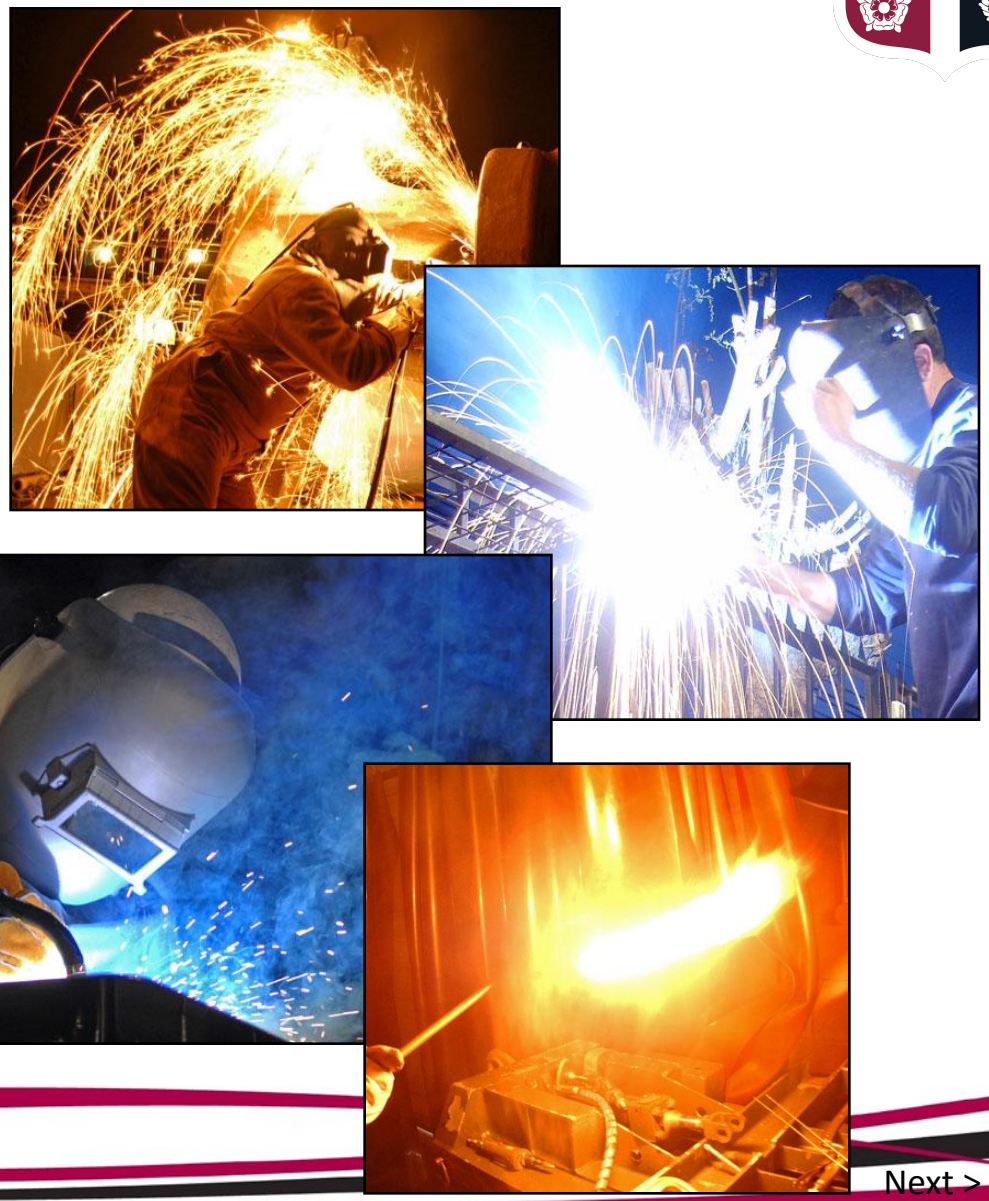
Oxyacetylene Gas Welding



The Dangers of Gas Welding

Here are some risks during welding:

- *Burns – caused by sparks and weld spatter*
- *Eye injuries – from looking at the bright flame*
- *Asphyxiation – from welding fumes*
- *Fire or explosion – if nearby materials are flammable*



Next >

PPE for Gas Welding

The following PPE should be worn when gas welding:

- *Shaded goggles for eye protection*
- *A mask to protect against fumes*
- *Flame resistant clothing for maximum body protection*
- *Flame resistant welding gauntlets*





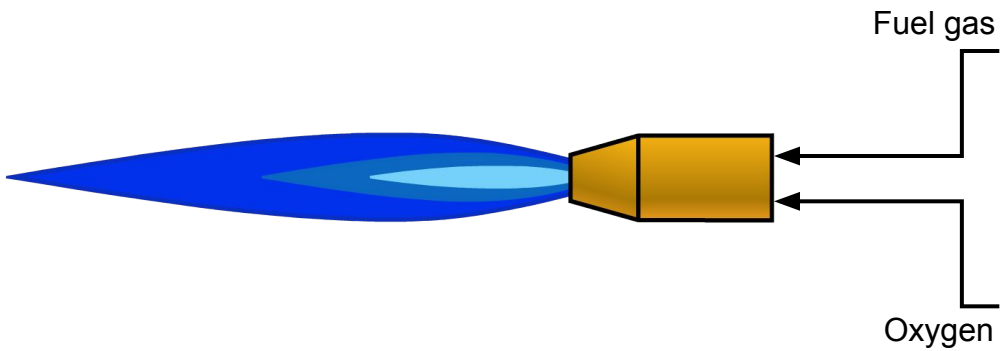




Next >

Gases for Gas Welding

To achieve a high welding temperature, the fuel gas is burned with pure oxygen.



The fuel gas may be hydrogen, propane or acetylene.

Fuel Gas	Flame Temperature
Hydrogen	2100 ˚C
Propane	2800 ˚C
Acetylene	3200 ˚C

The hottest flame is produced by a combination of oxygen and acetylene.

Next >

Question 1

Which of the following statements is true?

A) Gas welding is still used as a major welding process

B) A combination of oxygen and hydrogen produces the hottest flame

C) The oxygen hose is red and the fuel gas hose is green

D) Flux in the welding rod vaporises to displace air, preventing oxidation

Next >

Question 1

Which of the following statements is true?

A) Gas welding is still used as a major welding process

B) A combination of oxygen and hydrogen produces the hottest flame

C) The oxygen hose is red and the fuel gas hose is green



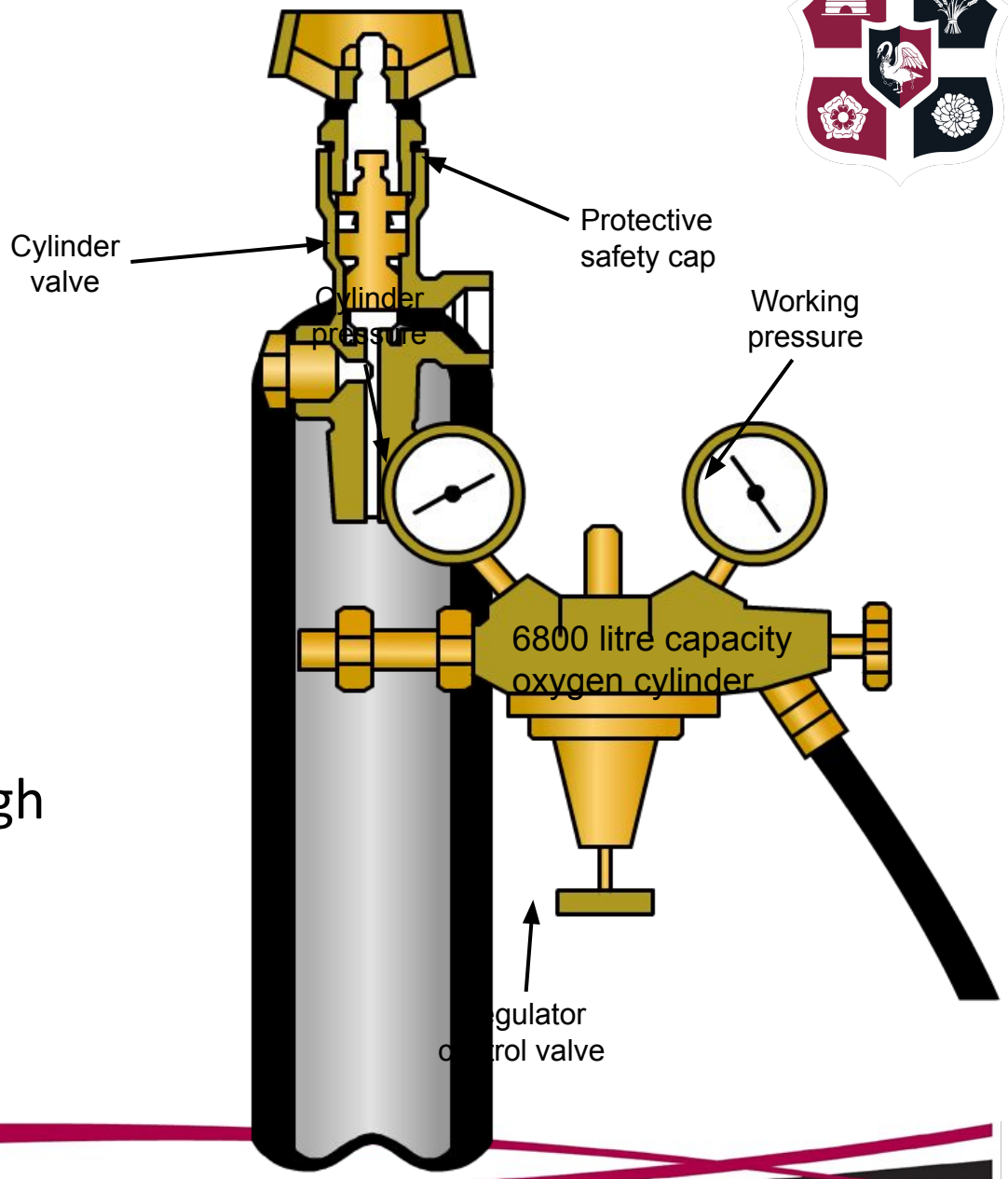
D) Flux in the welding rod vaporises to displace air, preventing oxidation

Oxygen Gas Storage

Oxygen is stored under high pressure in hollow steel cylinders.

The cylinder valve has a protective cap and should never be left exposed.

The regulator reduces the high cylinder pressure to a lower working pressure.



Next >

Oxygen Gas Dangers

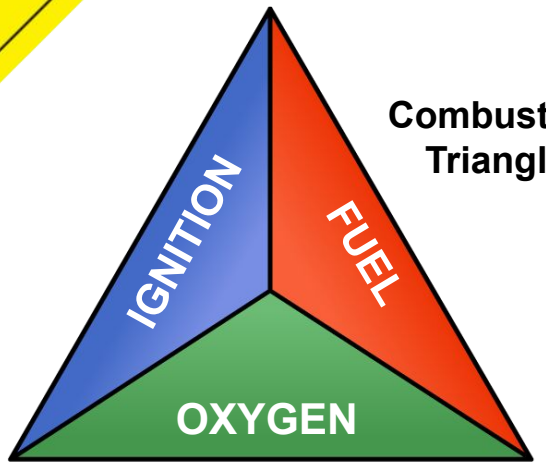
On its own, oxygen will not burn. It is an oxidiser, causing other materials to burn more vigorously.

Oxygen is one of three factors necessary for combustion.

Never use any hydrocarbons, such as oil or grease, near pure oxygen; they may spontaneously combust.



Warning sign for vehicles carrying oxygen



Combustion Triangle

Fuel + Oxygen + Ignition =



Next >

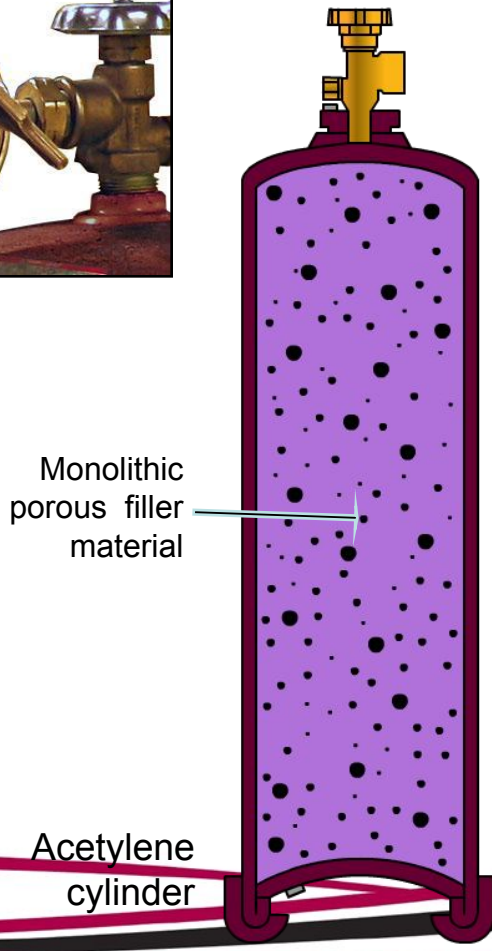
Acetylene Gas Storage

Acetylene gas is extremely unstable when pressurised above 1 bar.



For storage in cylinders, the gas is dissolved in acetone soaked into a porous filler material.

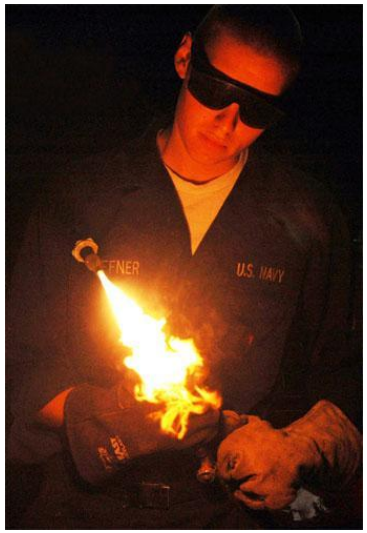
Cylinders should always be stored upright to prevent acetone leaks.



Next >

Acetylene Gas Dangers

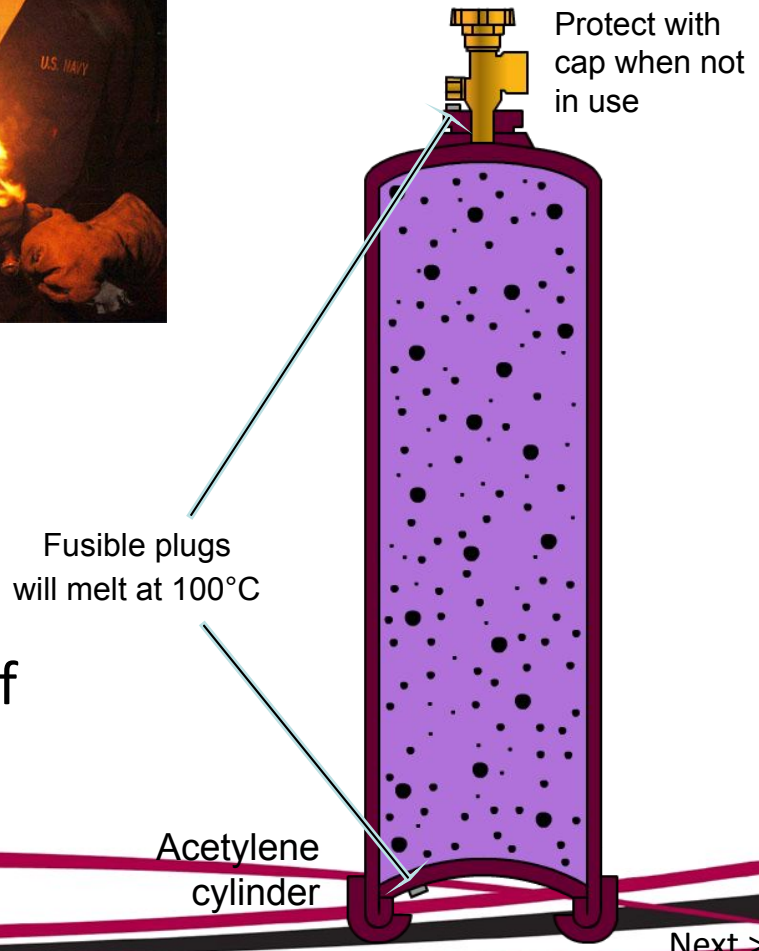
Acetylene burns in air with a yellow flame and black smoke.



It is shock sensitive and requires careful handling.

A fusible plug can melt to prevent fire causing an explosion.

Only open the cylinder valve ½ to ¾ of a turn, for a quick turn off in case of fire.



Next >

Question 2

What type of safety device is used on an acetylene cylinder?

A) A check valve

B) A cylinder valve

C) A fusible plug

D) A pressure regulator

Next >

Question 2

What type of safety device is used on an acetylene cylinder?

A) A check valve

B) A cylinder valve



C) A fusible plug

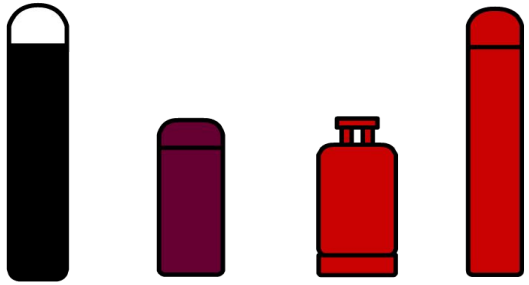
D) A pressure regulator

If there is a fire, the fusible plug on an acetylene cylinder will melt, preventing the cylinder from exploding.

Next >

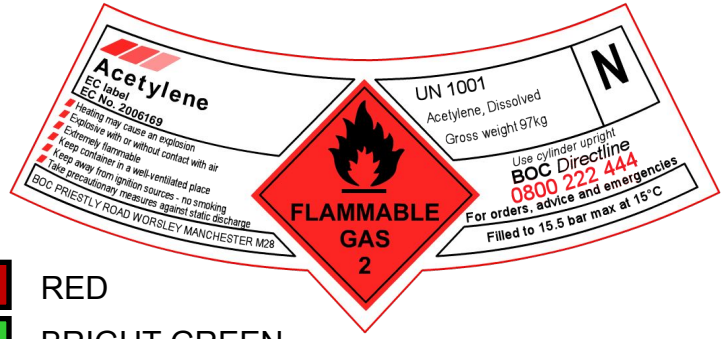
Gas Cylinder Markings

In Europe, cylinders are marked in accordance with standard EN 1089-3.



Oxygen | Dissolved acetylene | Propane | Hydrogen

Every cylinder must have a label with product and safety information.



Cylinder shoulders are also colour-coded to help identify the gas type.

Flammable	RED
Inert	BRIGHT GREEN
Oxygen	WHITE
Argon	DARK GREEN
Nitrogen	BLACK
Carbon dioxide	GREY
Helium	BROWN
Acetylene	MAROON

Next >

Question 3

In accordance with standard EN 1089-3, what is the shoulder colour of a cylinder filled with acetylene gas?

A) White

B) Maroon

C) Grey

D) Brown

Next >

Question 3

In accordance with standard EN 1089-3, what is the shoulder colour of a cylinder filled with acetylene gas?

A) White



B) Maroon

C) Grey

D) Brown

Gas Cylinder Safety

Cylinders must be stored with adequate ventilation and transported upright with regulators removed and safety caps fitted.



They should be secured to a cart or permanent structure.

Before use, cylinders should be inspected for damage.

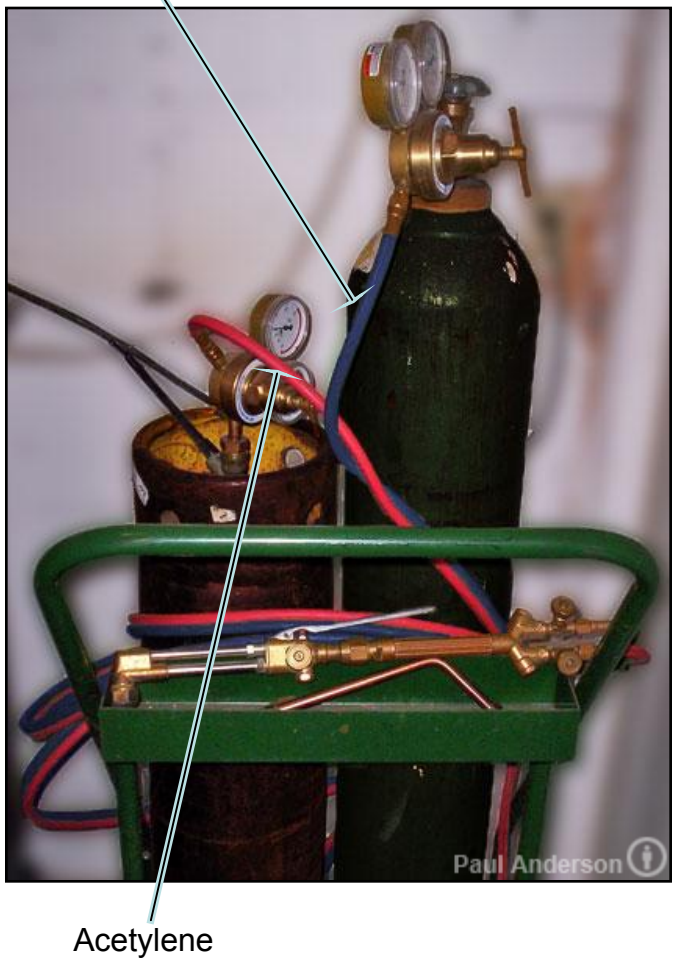


Next >

Gas Hoses and Connections

To avoid incorrect connection, gas fittings have different threads and the hoses are colour-coded.

- *Oxygen: right hand threads on blue hoses*
- *Acetylene: left hand threads on red or black hoses*



Next >

Gas Safety Devices

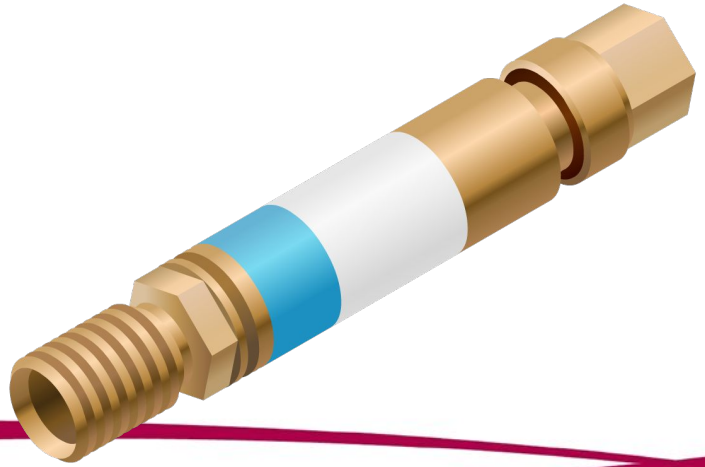
Check valves allow gas to flow in one direction only.

They are fitted to the welding torch.



Flashback arrestors prevent flames from reaching the gas in the cylinders.

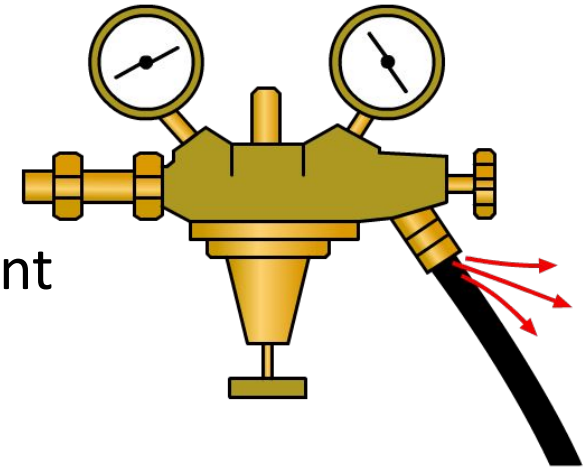
They are fitted to the pressure regulator.



Next >

Gas Welding Safety Rules

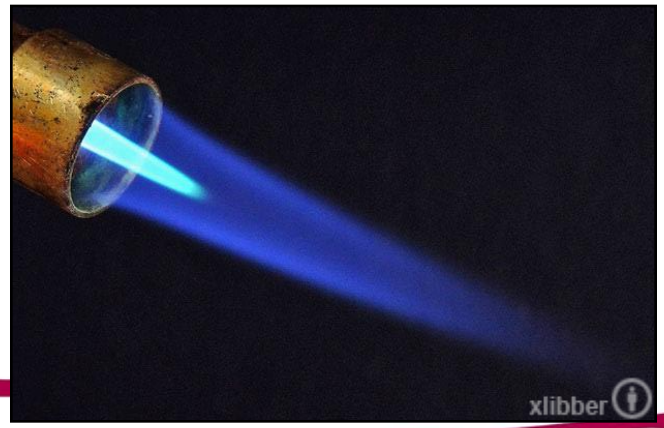
Inspect welding equipment for wear and leaks.





Open valves slowly to prevent damage from sudden high pressure.

Open and ignite the acetylene first. Then open and adjust the oxygen to obtain a neutral (blue) flame.



Position a suitable fire extinguisher close by.

Next >

Summary

In this presentation you have seen:

- *the gas welding process*
- *safety concerns associated with gas welding*

End